

APPLIED PHYSICS

UNIT I – QUANTUM PHYSICS

This unit introduces the fundamental concepts of **Quantum Physics**, which explain the behavior of matter and energy at atomic and subatomic levels. Classical physics fails to describe microscopic phenomena, and quantum theory provides accurate explanations through wave–particle duality, matter waves, uncertainty principle, and wave mechanics.

1. Wave–Particle Duality

Wave–particle duality states that every particle of matter and radiation exhibits both wave-like and particle-like properties.

- Light behaves as a wave in interference and diffraction.
- Light behaves as a particle (photon) in photoelectric effect.
- Similarly, electrons show particle nature as well as wave nature.

This dual nature is a fundamental concept of quantum mechanics.

2. de Broglie's Matter Waves

Louis de Broglie proposed that moving particles are associated with waves, called **matter waves**.

de Broglie Wavelength

The wavelength associated with a particle of momentum p is given by:

$$\lambda = h / p = h / mv$$

where:

- h = Planck's constant
 - m = mass of particle
 - v = velocity of particle
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3. Characteristics of Matter Waves

- Matter waves are not electromagnetic in nature.
 - They exist only when the particle is in motion.
 - Wavelength is inversely proportional to momentum.
 - For macroscopic objects, wavelength is very small and negligible.
 - For microscopic particles like electrons, wavelength is significant.
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4. Davisson and Germer's Experiment

This experiment experimentally verified the wave nature of electrons.

Principle

Electrons accelerated through a potential difference are diffracted by a crystal, similar to X-ray diffraction.

Observation

- A strong intensity peak was observed at certain angles.
- The measured wavelength matched de Broglie's wavelength.

Conclusion

Electrons exhibit wave nature, confirming de Broglie's hypothesis.

5. Heisenberg's Uncertainty Principle

It states that it is impossible to simultaneously measure the exact position and momentum of a particle.

Mathematical Form

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

where:

- Δx = uncertainty in position
- Δp = uncertainty in momentum

Significance

- It disproves deterministic nature of classical mechanics.

- It is significant only for microscopic particles.
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6. Schrödinger Time-Independent Wave Equation

Schrödinger developed a wave equation to describe the quantum behavior of particles.

One-Dimensional Time-Independent Equation

$$-(\hbar^2 / 2m) (d^2\psi / dx^2) + V(x)\psi = E\psi$$

where:

- ψ = wave function
 - $V(x)$ = potential energy
 - E = total energy
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7. Physical Significance of Wave Function

The wave function ψ itself has no direct physical meaning.

The probability of finding a particle is given by:

$$|\psi|^2 dx$$

Thus, ψ must be finite, continuous, and single-valued.

8. Characteristics of Wave Function

- ψ must be finite everywhere.
 - ψ must be single-valued.
 - ψ must be continuous.
 - ψ must be normalizable.
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9. Particle in a One-Dimensional Potential Box

Consider a particle confined in a one-dimensional box of length L with infinite potential walls.

Energy Eigenvalues

$$E_n = (n^2 h^2) / (8mL^2), n = 1, 2, 3, \dots$$

Features

- Energy is quantized.
 - Zero-point energy exists.
 - Energy levels increase as n^2 .
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10. Extension to Three-Dimensional Box

For a particle in a three-dimensional box of dimensions L_x, L_y, L_z , the energy is:

$$E_{n_x n_y n_z} = (h^2 / 8m) [(n_x^2 / L_x^2) + (n_y^2 / L_y^2) + (n_z^2 / L_z^2)]$$

where $n_x, n_y, n_z = 1, 2, 3, \dots$

UNIT II – BAND THEORY OF SOLIDS

This unit explains the **electronic behavior of solids** using band theory. It describes how electrons behave in solids, the formation of energy bands, and how materials are classified into conductors, semiconductors, and insulators. These concepts are fundamental to solid-state electronics and semiconductor devices.

1. Free Electron Theory of Metals

Free electron theory explains the electrical properties of metals by assuming that valence electrons move freely inside the metal.

1.1 Classical Free Electron Theory (Drude Model)

Assumptions:

- Electrons behave like free particles inside the metal
- Positive ions form a fixed lattice
- Electron–electron interaction is neglected

Merits:

- Explains electrical conductivity
- Explains thermal conductivity

Limitations:

- Cannot explain temperature dependence of conductivity accurately
 - Fails to explain specific heat of electrons
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1.2 Quantum Free Electron Theory (Sommerfeld Model)

This theory applies **quantum mechanics** to free electrons.

Features:

- Electrons obey Fermi–Dirac statistics
 - Energy levels are quantized
 - Explains electrical and thermal properties better than classical theory
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2. Fermi Level

The **Fermi level** is defined as the maximum energy level occupied by electrons at absolute zero temperature (0 K).

Significance:

- Determines electrical conductivity
 - Position of Fermi level distinguishes conductors, semiconductors, and insulators
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3. Density of States

Density of states (DOS) represents the number of available electron energy states per unit energy range.

Importance:

- Helps calculate electron concentration
- Used in analyzing electrical and optical properties

4. Periodic Potential in Solids

In a crystal, atoms are arranged periodically, producing a **periodic potential field** experienced by electrons.

This periodic potential leads to the formation of **allowed and forbidden energy bands**.

5. Bloch's Theorem

Bloch's theorem states that the wave function of an electron in a periodic potential can be written as:

$$\psi(x) = u(x) e^{ikx}$$

where $u(x)$ has the same periodicity as the crystal lattice.

Significance:

- Forms the basis of band theory
 - Explains electron motion in crystals
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6. Kronig–Penney Model

The Kronig–Penney model is a one-dimensional model that explains the formation of energy bands using a periodic potential.

Results:

- Predicts allowed energy bands
- Predicts forbidden energy gaps

This model provides a simple mathematical explanation of band structure.

7. E–k Diagram

The **E–k diagram** represents the relationship between energy (E) and wave vector (k).

Features:

- Shows allowed and forbidden bands
 - Slope gives electron velocity
 - Curvature indicates effective mass
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8. Effective Mass of Electron

Effective mass is the apparent mass of an electron in a crystal lattice.

Expression:

$$1 / m^* = (1 / \hbar^2) (d^2E / dk^2)$$

Importance:

- Determines mobility of charge carriers
 - Can be smaller or larger than free electron mass
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9. Origin of Energy Bands in Solids

When isolated atoms come together to form a solid:

- Atomic energy levels split
 - Closely spaced levels form energy bands
 - Gaps between bands are forbidden energy gaps
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10. Classification of Materials Based on Band Theory

10.1 Conductors

- Valence band overlaps with conduction band
- No energy gap
- High electrical conductivity

Example: Copper, Aluminum

10.2 Semiconductors

- Small energy gap (~ 1 eV)
- Conductivity increases with temperature

Example: Silicon, Germanium

10.3 Insulators

- Large energy gap (> 3 eV)
- Very low electrical conductivity

Example: Glass, Rubber

UNIT III – SEMICONDUCTOR PHYSICS

This unit deals with the fundamental principles of **semiconductor physics**, explaining intrinsic and extrinsic semiconductors, carrier concentration, Fermi level behavior, charge transport mechanisms, PN junction formation, diode characteristics, and the Hall effect. These concepts form the basis of modern electronic devices.

1. Semiconductors

Semiconductors are materials whose electrical conductivity lies between that of conductors and insulators. Their conductivity can be controlled by temperature and impurity addition.

Examples: Silicon (Si), Germanium (Ge)

2. Intrinsic Semiconductor

An intrinsic semiconductor is a pure semiconductor without any impurity atoms.

Characteristics

- Equal number of electrons and holes
- Conductivity increases with temperature

- Fermi level lies near the middle of the energy gap

Carrier Concentration in Intrinsic Semiconductor

The number of electrons (n) is equal to the number of holes (p).

$$n = p = n_i$$

where n_i is the intrinsic carrier concentration.

3. Extrinsic Semiconductor

An extrinsic semiconductor is formed by adding controlled impurities to an intrinsic semiconductor. This process is called **doping**.

Types of Extrinsic Semiconductors

- n-type semiconductor
 - p-type semiconductor
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4. n-Type Semiconductor

n-type semiconductors are formed by doping with pentavalent impurities such as Phosphorus or Arsenic.

Donor Concentration

- Donor atoms donate free electrons
- Electrons are majority carriers
- Holes are minority carriers

$$n \approx N^D$$

5. p-Type Semiconductor

p-type semiconductors are formed by doping with trivalent impurities such as Boron or Gallium.

Acceptor Concentration

- Acceptor atoms create holes

- Holes are majority carriers
- Electrons are minority carriers

$$p \approx N^A$$

6. Variation of Fermi Level

With Temperature

- In intrinsic semiconductors, Fermi level shifts towards the center with increase in temperature
- In extrinsic semiconductors, it moves closer to intrinsic level at high temperatures

With Doping Concentration

- In n-type: Fermi level moves towards conduction band
 - In p-type: Fermi level moves towards valence band
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7. Carrier Transport Mechanisms

Charge carriers move in semiconductors by two mechanisms:

7.1 Drift

Drift is the movement of charge carriers under the influence of an electric field.

Drift Velocity:

$$v_d = \mu E$$

where μ is mobility and E is electric field.

7.2 Diffusion

Diffusion is the movement of carriers from a region of high concentration to a region of low concentration.

Diffusion occurs due to concentration gradient.

8. Formation of PN Junction

A PN junction is formed when p-type and n-type semiconductors are joined together.

Key Features

- Formation of depletion region
- Built-in potential barrier
- No free charge carriers in depletion region

9. V–I Characteristics of PN Junction Diode

Forward Bias

- Depletion width decreases
- Current increases rapidly after cut-in voltage

Reverse Bias

- Depletion width increases
- Small reverse saturation current flows

10. Hall Effect

The Hall effect is the generation of a transverse voltage when a current-carrying conductor or semiconductor is placed in a magnetic field.

Hall Voltage

$$V_H = (IB) / (nqt)$$

where I = current, B = magnetic field, n = carrier concentration, q = charge, t = thickness.

11. Applications of Hall Effect

- Determination of type of semiconductor

- Measurement of carrier concentration
- Measurement of mobility
- Magnetic field sensing

UNIT IV – OPTOELECTRONIC, DIELECTRIC AND DISPLAY DEVICES

This unit focuses on **optoelectronic devices**, **dielectric materials**, and **display technologies**, which play a vital role in modern electronics, communication systems, sensors, and display applications.

PART A: OPTOELECTRONIC DEVICES

Optoelectronic devices are electronic devices that either emit or detect light using semiconductor materials.

1. Direct and Indirect Band Gap Semiconductors

Direct Band Gap Semiconductors

In direct band gap semiconductors, the minimum of the conduction band and maximum of the valence band occur at the same momentum value.

- Efficient light emission
- Used in LEDs and laser diodes

Examples: GaAs, InP

Indirect Band Gap Semiconductors

In indirect band gap semiconductors, conduction band minimum and valence band maximum occur at different momentum values.

- Poor light emission
- Used mainly in electronic devices

Examples: Silicon, Germanium

2. Light Emitting Diode (LED)

Construction

An LED consists of a heavily doped p–n junction made from direct band gap semiconductors.

Working Principle

- When forward biased, electrons recombine with holes
- Energy is released in the form of light

Applications

- Display indicators
 - Traffic signals
 - Optical communication
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3. Laser Diode

Construction

A laser diode is a heavily doped p–n junction with an optical cavity formed by reflective surfaces.

Working Principle

- Operates on stimulated emission
- Produces coherent, monochromatic, and directional light

Applications

- Optical fiber communication
 - Barcode scanners
 - CD/DVD drives
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4. Photodiode

A photodiode is a light-sensitive p–n junction device operated in reverse bias.

Working Principle

- Incident light generates electron–hole pairs

- Reverse current increases with light intensity

Applications

- Light sensors
 - Optical receivers
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5. PIN Diode

A PIN diode consists of a p-region, intrinsic layer, and n-region.

Advantages

- Increased depletion width
- Higher sensitivity
- Faster response

Applications

- Optical communication
 - RF switches
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6. Avalanche Photodiode (APD)

An APD operates at high reverse bias and provides internal gain due to avalanche multiplication.

Features

- High sensitivity
- Low noise

Applications

- Fiber optic systems
 - Low-light detection
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7. Solar Cell

A solar cell converts solar energy into electrical energy using photovoltaic effect.

Working Principle

- Light generates charge carriers
- Built-in electric field separates carriers

Applications

- Renewable energy systems
 - Satellites
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PART B: DIELECTRIC AND DISPLAY MATERIALS

8. Dielectric Materials

Dielectrics are insulating materials that become polarized in the presence of an electric field.

Applications

- Capacitors
 - Electrical insulation
 - Energy storage
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9. Polarization

Polarization is the alignment of dipoles in a dielectric material under an electric field.

Types of Polarization (Qualitative)

(a) Ferroelectric Materials

- Exhibit spontaneous polarization
- Polarization can be reversed by electric field

Examples: Barium Titanate

(b) Piezoelectric Materials

- Generate electric charge under mechanical stress

Examples: Quartz

(c) Pyroelectric Materials

- Generate electric charge due to temperature change

Examples: Tourmaline

10. Luminescence

Luminescence is the emission of light by a substance not resulting from heat.

10.1 Photoluminescence

Light emission due to absorption of photons.

10.2 Cathodoluminescence

Light emission due to electron bombardment.

10.3 Electroluminescence

Light emission due to electric field or current.

10.4 Injection Electroluminescence

Occurs due to recombination of charge carriers across a p–n junction.

11. Display Devices

11.1 Plasma Display

Working Principle

- Uses ionized gas cells
 - Produces ultraviolet light that excites phosphors
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11.2 Liquid Crystal Display (LCD)

Working Principle

- Uses liquid crystals to control light transmission
 - Low power consumption
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11.3 Numerical Displays

Numerical displays use seven-segment LEDs or LCDs to display digits.

UNIT V – FIBER OPTICS AND QUANTUM INFORMATION SCIENCE

This unit covers **optical fiber technology**, which is the backbone of modern communication systems, and **quantum information science**, an emerging field that applies quantum mechanics to information processing and computation.

PART A: FIBER OPTICS

1. Introduction to Optical Fiber

An optical fiber is a thin, flexible, transparent dielectric medium used to transmit light signals over long distances with very low loss. It works on the principle of **total internal reflection**.

Advantages:

- High bandwidth
- Low transmission loss
- Immunity to electromagnetic interference
- Secure communication

2. Construction of an Optical Fiber

An optical fiber consists of three main parts:

- **Core:** Central region that carries light signals
 - **Cladding:** Surrounding layer with lower refractive index
 - **Protective Jacket:** Outer coating for mechanical protection
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3. Working Principle of Optical Fiber

Optical fiber operates based on **total internal reflection**.

- Light enters the core at a suitable angle
 - It undergoes repeated total internal reflections at the core–cladding interface
 - Light propagates through the fiber with minimal loss
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4. Acceptance Angle and Numerical Aperture

Acceptance Angle

The maximum angle at which light can enter the fiber and still be guided through the core.

Numerical Aperture (NA)

Numerical aperture determines the light-gathering ability of the fiber.

Expression:

$$NA = \sqrt{n_1^2 - n_2^2}$$

where n_1 = refractive index of core, n_2 = refractive index of cladding.

5. Types of Optical Fibers

Based on mode of propagation:

- Single mode fibers

- Multimode fibers

Based on refractive index profile:

- Step index fibers
 - Graded index fibers
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6. Step Index and Graded Index Fibers

Step Index Fiber

- Sudden change in refractive index
- Higher modal dispersion
- Used for short-distance communication

Graded Index Fiber

- Gradual change in refractive index
 - Reduced dispersion
 - Used for medium-distance communication
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7. Attenuation in Optical Fiber

Attenuation is the reduction of signal strength during transmission.

Causes:

- Absorption loss
 - Scattering loss
 - Bending loss
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8. Applications of Optical Fiber

- Optical fiber communication systems
- Internet and telecommunication
- Medical endoscopy

- Sensors and instrumentation
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9. Optical Fiber Communication System

An optical fiber communication system consists of:

- Optical transmitter (LED or laser)
- Optical fiber channel
- Optical receiver (photodiode)

It enables high-speed and long-distance data transmission.

PART B: QUANTUM INFORMATION SCIENCE

10. Introduction to Quantum Information Science

Quantum information science applies the principles of quantum mechanics to information processing, communication, and computation.

11. Quantum Mechanical Bit (Qubit)

A **qubit** is the basic unit of quantum information.

Unlike a classical bit (0 or 1), a qubit can exist in a superposition of both states.

12. Superposition and Entanglement

Superposition

A qubit can exist simultaneously in states $|0\rangle$ and $|1\rangle$.

Entanglement

Entanglement is a quantum phenomenon where two or more qubits become correlated such that the state of one depends on the other.

13. Dirac Notation and Quantum States

Quantum states are represented using **Dirac notation (bra-ket notation)**.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are probability amplitudes.

14. Hilbert Space and Quantum State Vector

Quantum states exist in a complex vector space called **Hilbert space**.

The quantum state vector represents the state of a quantum system in this space.

15. Bloch Sphere Representation of Qubit

The Bloch sphere is a geometrical representation of a qubit.

- Any pure qubit state lies on the surface of the sphere
 - Provides visualization of qubit states
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16. Quantum Logic Gates

Quantum logic gates manipulate qubits through unitary operations.

16.1 Single Qubit Gates

- Pauli-X gate
 - Pauli-Y gate
 - Pauli-Z gate
 - Hadamard gate
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16.2 Multiple Qubit Gates

- CNOT gate
 - Controlled-Z gate
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17. Applications of Quantum Information Science

- Quantum computing
- Quantum cryptography
- Quantum teleportation
- Secure communication